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Peer Review File

Machine learning for comprehensive interaction modelling improves disease risk prediction in the UK Biobank

Corresponding Author: Dr Heli Julkunen

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The proposed method shows moderate prediction power improvement in time-to-events data that leverage low-rank modelling of interaction effects. This work builds on past work that uses non-linear risk models for disease prediction (e.g. Placido et al. 2023 Nature Medicine) and low-rank modelling of health data (e.g. Jiang et al. 2023 Nature Genetics). The analyses are comprehensive: the methodology is well written and tested on a comprehensive set of predictors separated into four scenarios; the comparison with QRISK3 and the choice of C-index and NRI increase the translational impact of the work. For prediction models, if the authors could show the power in external data sets, it would be more convincing. I have following main comments:

1. External validation. UK Biobank has individuals who have kinship that is known to cause inflation for prediction models (Ellis 2024 AJHG). Since the model is fitting many predictors, there is a chance that the model is capturing subtle population structures that cross-validation could not remove. One solution is finding an external validation cohort; the other solution is making sure individuals in the training and testing don't have kinship (i.e. the same family are always in the training OR testing).
2. Providing an additional metric to evaluate the added value of interaction terms. The c-index improvement is small and not easily interpretable (i.e. improvement from 0.6 to 0.7 has a very different interpretation than 0.7 to 0.8; even the same C-index with different incidence rate is not comparable). Is it possible to quantify the contribution of prediction power by the interaction terms (analogous to R2)? Previous study has shown that for traits with many small effects, non-additive variance is small (e.g. in genetics, Palmer et al. 2023 Science and Hill et al. 2008 PLoS Genetics). It would be interesting to show the amount of prediction power added through the interaction components compared to the linear model (e.g. Shapley value); increment of C-index doesn’t provide this information as the increment is not comparable to the C-index of the linear model. This value might not be large, but it is still interesting to report and I don’t think this will reduce the contribution of this work.
3. The proposed survivalFM model uses L2 regularization in parameter estimation, which generates non-zero coefficient estimates for all marginal and interaction terms. Considering the sparsity of the interaction effect matrices under several scenarios (e.g., Supplementary Figure 7-8, 11), would using L1 penalty, or a combination of L1 and L2, further improve the model performance as well as the interpretability of the model?

Specifically:
Fig 1b: the expression of “time to event ~ linear terms” is not Cox model; this is a different type of survival model that directly models the time to event as linear combination of predictors (Cox and Oakes 1983).

Line 150: Based on Supplementary Figure 6, the continuous NRI among patients with events tends to be negative for many outcomes across different models, despite that the overall NRI is positive in general. Given that it is important to correctly identify high-risk patients who experience events, it will be helpful to summarize the NRI results separately for events and non-events in the main manuscript and provide interpretations accordingly.

Line 360 – couple of questions related to model fitting: how does the model decide the number of latent factors? (connected to main comment) Is there a possibility of overfitting the subtle relatedness between training and testing through e.g. kinship? (should we ask for an external validation cohort)?

Line 495: How is the number of latent factors chosen, and how does the number of latent factors affect the predictive performance?

Line 363: thanks for providing the detailed derivation of gradient in equation 5-10; connected to line 488 – what are the procedures for selecting hyperparameters? Line 494 mentioned grid search but did not specify what are the criteria used (is it based on cross validation of likelihood? Prediction C-index?) (edit: this is explained in line 129-130)

Line 107 – 109: Please provide more details on how prevalent cases were excluded. How did you exclude the prevalent diseases listed in ST2? Some of the diseases don't have primary care data available.

Line 127: for some analysis (group 1, 2, and 4) it is also possible to run the analysis including all pairwise interaction terms. It is worth including a model with pairwise interaction terms as a comparison. This will potentially show under what predictor structure SurvivalFM shows best prediction power, in additional to sample size which is studied.

Line 212: connected to the main point – from the figure the improvement is small. It might be interesting to quantify what are the additional contributions of interaction for each of data modality.

Line 247: Given the guideline recommended 10% risk threshold for QRISK3, for each model, it might be helpful to also report the observed risk of CVD event at 10 years (as well as other key time points) based on Kaplan-Meier analysis for patients with a predicted risk >=10% and those with a predicted risk <10%. This can provide an interpretable and clinically relevant summary of discrimination performance for each model which complement C-index and NRI reported.

Supplementary Figure 15: Figure legend should refer to a) and b) rather than c) and d).

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)
The survivalFM method is implemented in an R package, which can be a useful and user-friendly tool for interaction modeling in disease risk prediction. The Github page contains a README file with instructions on installation and example usage of the R package. I was able to install the R package and run the example code without issues.

Reviewer #3

(Remarks to the Author)
The paper introduces survivalFM, a machine learning extension of the Cox proportional hazards model that incorporates pairwise interactions using a low-rank factorisation approach (from factorisation machines) to predict time-to-event outcomes. The authours evaluate survivalFM on 10-year disease outcomes using data from the UK Biobank, demonstrating improvements over traditional Cox models across multiple diseases and data modalities. The authours also specifically compare it to QRISK3 for cardiovascular risk prediction. The method shows particular strength in improving continuous NRI, while maintaining a level of interpretability.

The paper is well-written and demonstrates the potential of survivalFM for time-to-event prediction, but a few points would benefit from further clarification:

Interpretability of Latent Vectors: While it is true that the contribution of interaction terms to f(x) can be explicitly seen, could the authours clarify how the interaction coefficients derived from latent vectors in survivalFM differ in their clinical and statistical interpretation from traditional Cox models? In traditional Cox models, \beta_{ij} directly quantifies the strength and direction of the interaction between two covariates x_i and x_j, which is straightforward to interpret in a clinical context. However, in survivalFM, \langle p_i, p_j \rangle, the interaction derived from latent vectors, lacks the same direct interpretability, as it represents a more abstract relationship. How should clinicians understand these latent interactions in practice? In Figure 5, the heatmap shows an interaction coefficient between mutually exclusive categories (e.g., heavy smoker:moderate smoker). How should this interaction be interpreted, and is it meaningful? Are there other unexpected interaction terms that might need further explanation?
Selection of Latent Dimension (k): The authours use k=10 for the latent space dimension across all models. Could the rationale for this choice be explained? What guidelines could be provided for selecting k in different datasets or scenarios?
List of Disease Outcomes: Could the authours explicitly list the 10 disease outcomes in the Methods section (I see they are buried in Supplementary Table 2)? This would help readers assess the diversity of outcomes and the fraction of improvements across scenarios.
Model Composition: Why was a combined model using standard risk factors, clinical measures, metabolomics, and polygenic risk scores not included?

Thank you for your consideration.

(Remarks on code availability)

Reviewer #4

(Remarks to the Author)
Machine learning for comprehensive interaction modelling improves disease risk prediction in the UK Biobank

Heli Julkunen and Juho Rousu

The authors propose a model for all interactions of a Cox proportional hazards regression by projecting into a space of lower dimensions. Of course, this is also applicable to all linear models. As an example, the manuscript describes an application to the huge UK Biomank database combining clinical and genomic risk factors for a variety of diseases.

Comments

The present manuscript feels like an advertisement for the author's software package survivalFM. This manuscript would be an ideal candidate for a journal of statistical software. There is a lot of text devoted to the capabilities of survivalFM. If this is available in R then there are other, more suitable outlets for this manuscript

There is no strong connection to machine learning and this is not explained in any depth.

There should be some guidance on how the dimensions of the p-parameters are chosen. In principal components analysis, for example, there is a scree plot to help us decide the appropriate dimension of the underlying latent measure. Similarly, there is no interpretation of the estimated p-vector.
For a given dimension of p, there is a good explanation (on page 10) on how p is estimated. The choice of the two lambda parameters in (7) is very similar to the way penalized linear regression chooses the midground between lasso and ridge regression estimation.

The frequent use of the word "prediction" is misleading. The manuscript is not predicting anything, but rather, seeking to build a mathematical model to explain what has already been observed.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors addressed my concerns well; I have the following additional (minor) comments:
1. Based on the updated results, the absolute gains in C-index and R squared are small in general from modeling interaction effects, while the predictive improvement mainly lies in NRI. Could the authors further comment on how this result aligns with previous studies modeling interaction terms in risk models, in terms of small non-additive effects compared to linear terms? The authors are suggested to clarify that including interaction terms provides modest improvement in prediction performance, which aligns with previous studies. The advantage in improving individual-level risk prediction despite small improvement at population level can be potentially emphasized.
2. Figure 2 and line 164-166: since continuous NRI is not a proportion, might consider not using the percentage (as NRI can go up to 200%).
3. Figure 4: can you clarify if the grey line is Cox model with or without interaction terms? It is a fair comparison between Cox model with interaction term and survivalFM as survivalFM might require less samples to capture the interaction information. If the grey line is cox model with only main effect, it worth also putting the Cox model with all pairwise interactions in the figure.
4. I noticed the first two numbers (and the corresponding lower bound) in row 1 of Table 1 are exactly the same.
5. Line 160 and Table S4: absolute R2 values range from 24.9% to 95.0%, which seems very high. Is this due to the specific properties of Royston's R2?
6. Line 237 and Supplementary Figure 10: Would be better to say R > 0.99 instead of R = 1, which implies a perfect linear relationship.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
Thank you for your efforts in revising the manuscript. The updates have notably improved its clarity and rigour, and I now find the manuscript suitable for publication.

(Remarks on code availability)
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Reviewer #4

(Remarks to the Author)

(Remarks on code availability)

NCOMMS-24-46967-T Point-by-point response to reviewer comments

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The proposed method shows moderate prediction power improvement in time-to-events data that leverage low-rank modelling of interaction effects. This work builds on past work that uses non-linear risk models for disease prediction (e.g. Placido et al. 2023 Nature Medicine) and low-rank modelling of health data (e.g. Jiang et al. 2023 Nature Genetics). The analyses are comprehensive: the methodology is well written and tested on a comprehensive set of predictors separated into four scenarios; the comparison with QRISK3 and the choice of C-index and NRI increase the translational impact of the work. For prediction models, if the authors could show the power in external data sets, it would be more convincing. I have following main comments:

1. External validation. UK Biobank has individuals who have kinship that is known to cause inflation for prediction models (Ellis 2024 AJHG). Since the model is fitting many predictors, there is a chance that the model is capturing subtle population structures that cross-validation could not remove. One solution is finding an external validation cohort; the other solution is making sure individuals in the training and testing don't have kinship (i.e. the same family are always in the training OR testing).

Our response:

We thank the reviewer for the insightful comments and valuable suggestions. We have now implemented two notable measures to address these concerns:

1. First, rather than relying on nested cross-validation procedures within the UK Biobank, we now train all the prediction models using data from individuals recruited in England and Wales, and test them on individuals recruited in Scotland. This aligns with the approaches adopted in several other UK Biobank prediction studies, including recent work by Zheng et al. (*Nature Communications*, 2024), Argentieri et al. (*Nature Medicine*, 2025), and earlier work by Ganna et al. (*The Lancet*, 2015). As noted in prior research (e.g., McCartney et al., *Journal of Public Health*, 2015), Scotland has been observed to exhibit demographic and health-related differences compared to England and Wales, enhancing the stringency of our model evaluation. This regional separation can be considered as a proxy for external validation and helps to reduce the influence of subtle population structures or relatedness on prediction accuracy.

   A key limitation of this updated approach is that the number of individuals recruited in Scotland is considerably smaller than the full UK Biobank dataset, leading to reduced statistical power to detect small improvements in predictive performance. Consequently, confidence intervals for performance metrics are wider with this new model evaluation setting. Additionally, to ensure a sufficient number of events in the test set, we had to combine the previously separate endpoints of alcoholic liver disease and liver fibrosis and cirrhosis along with other chronic liver disease outcomes into a single liver disease endpoint. While the smaller test set inevitably reduces statistical power, we believe this refinement improves the robustness of our model evaluation.

2. Second, to further mitigate the potential for kinship-induced inflation, we performed an additional analysis in which we excluded from the Scotland test set all individuals who shared kinship with any member of the England and Wales training set (defined as third-degree or closer relatives, with a kinship coefficient ≥ 0.0442). Results from this analysis are shown in Supplementary Figures 6-7 and discussed on lines 190-198 in the revised manuscript. Under this stricter criterion, we observed that improvements in predictive performance remained stable, with only minor numerical variation and no meaningful reduction in predictive performance. This finding suggests that the model's predictive performance is robust and not driven by inclusion of related individuals.

While an external validation cohort would be ideal, identifying another dataset with a sufficiently similar set of predictor variables is challenging. Publicly available datasets or cohorts with the same biomarker panels, especially the complete panel of metabolites and detailed clinical chemistry measures, are scarce and access can take a long time to obtain. Additionally, ensuring a sufficiently large sample size with these exact features further limits the availability of appropriate external datasets. However, we believe that the more rigorous internal validation measures incorporated in the revised manuscript, i.e. using the subcohort recruited in Scotland for testing the models and performing additional analyses excluding genetically related individuals, effectively address the reviewer's concerns regarding kinship and generalizability. These approaches are also consistent with methodologies adopted in other recent UK Biobank prediction studies.

2. Providing an additional metric to evaluate the added value of interaction terms. The c-index improvement is small and not easily interpretable (i.e. improvement from 0.6 to 0.7 has a very different interpretation than 0.7 to 0.8; even the same C-index with different incidence rate is not comparable). Is it possible to quantify the contribution of prediction power by the interaction terms (analogous to R2)? Previous study has shown that for traits with many small effects, non-additive variance is small (e.g. in genetics, Palmer et al. 2023 Science and Hill et al. 2008 PLoS Genetics). It would be interesting to show the amount of prediction power added through the interaction components compared to the linear model (e.g. Shapley value); increment of C-index doesn't provide this information as the increment is not comparable to the C-index of the linear model. This value might not be large, but it is still interesting to report and I don't think this will reduce the contribution of this work.

Our response:

We thank the reviewer for this great suggestion. To better quantify the contribution of the interaction terms and provide a more interpretable measure of their added predictive value, we have now included an incremental R^2 measure (ΔR^2), based on a measure of explained variation for survival models introduced by Royston et al. (*Statistics in Medicine,* 2006). Royston's R^2 extends the concept of explained variation from linear regression to survival outcomes, providing a measure of overall model fit. By comparing our survivalFM model with the comprehensive interaction terms to the standard Cox regression model, we assess the additional explained variation attributable to the interaction effects. In the revised manuscript, we now report both the incremental R^2 and incremental C-index (Figure 1, Table 1, Supplementary Table 4) and discuss these results on lines 153-160 and 315-322. We believe this addition effectively addresses the reviewer's concern and enhances the interpretation of our results.

3. The proposed survivalFM model uses L2 regularization in parameter estimation, which generates non-zero coefficient estimates for all marginal and interaction terms. Considering the sparsity of the interaction effect matrices under several scenarios (e.g., Supplementary Figure 7-8, 11), would using L1 penalty, or a combination of L1 and L2, further improve the model performance as well as the interpretability of the model?

Our response:

We thank the reviewer for this suggestion. As noted by the reviewer, the current implementation of the survivalFM model employs L2 regularization, which does not enforce sparsity. However, since the regularization is applied to the matrix **P**, which contains the latent factor vectors $\mathbf{p}_i$ (Eq. (7) in Methods), introducing L1 regularization would primarily promote sparsity in the individual factor vectors. However, this sparsity would not necessarily translate to sparsity in the pairwise inner products that define the interaction effects. This is because the remaining non-zero elements in the latent vectors could still result in non-zero inner products.

However, it is non-trivial to achieve a sparsity in the interactions as this would require making some of the factor vectors orthogonal (to achieve zero inner product) but retaining some of them non-zero. This requires special regularization techniques to be developed (c.f. related work by Atarashi, K., Oyama, S. and Kurihara, M., 2021. Factorization machines with regularization for sparse feature interactions. Journal of Machine Learning Research, 22(153), pp.1-50.) which we see as an interesting future direction. We have now added a discussion of this point to the revised manuscript (lines 406-411), highlighting it as an interesting area for future research.

Specifically:
Fig 1b: the expression of "time to event ~ linear terms" is not Cox model; this is a different type of survival model that directly models the time to event as linear combination of predictors (Cox and Oakes 1983).

Our response: We thank the reviewer for pointing out the ambiguity in the notation used in Figure 1b. The expression "time to event ~ " was intended to reflect the typical R syntax for model specification but indeed does not accurately represent the Cox proportional hazards model. We have now revised Figure 1 and the corresponding caption to explicitly indicate that the model is a Cox proportional hazards model using standard mathematical notation.

Line 150: Based on Supplementary Figure 6, the continuous NRI among patients with events tends to be negative for many outcomes across different models, despite that the overall NRI is positive in general. Given that it is important to correctly identify high-risk patients who experience events, it will be helpful to summarize the NRI results separately for events and non-events in the main manuscript and provide interpretations accordingly.

Our response: We appreciate this observation and have now addressed it by reporting the continuous net reclassification improvement (NRI) separately for individuals with events (NRI_{events}) and those without events ($NRI_{non\text{-}events}$) in the revised manuscript (Figure 2, lines 177-189).

Line 360 – couple of questions related to model fitting: how does the model decide the number of latent factors? (connected to main comment) Is there a possibility of overfitting the subtle relatedness between training and testing through e.g. kinship? (should we ask for an external validation cohort)?

Our response:

Regarding the number of latent factors, please see our response to the question below, where we address this aspect in detail.

To assess the potential inflation of predictive performance due to kinship-related overfitting, we have now conducted an additional analysis in which we excluded from the Scotland test set all individuals who shared kinship with any member in the England and Wales training set. We defined relatedness as third-degree or closer relatives, using a kinship coefficient threshold of ≥ 0.0442. Under this more stringent exclusion criterion, we found that the model's predictive performance in the test set remained stable, with only minor numerical fluctuations and no meaningful reduction in accuracy. These results indicate that the model's predictive ability is not inflated by the inclusion of related individuals in the test set. Results from this analysis are shown in Supplementary Figures 6-7 and discussed on lines 190-198 in the revised manuscript.

Line 495: How is the number of latent factors chosen, and how does the number of latent factors affect the predictive performance?

Our response:

The number of latent factors (rank *k*) is a central hyperparameter in factorization-based models, as it determines the dimensionality of the latent space representing feature interactions. A larger *k* allows for greater expressiveness, while a smaller *k* enforces a more compact representation. While cross-validation is one approach for tuning *k*, it becomes computationally prohibitive for large datasets. To manage the computational complexity, we adopted a pragmatic approach of selecting a moderate *k*=10, with L2 regularization applied separately to the latent factors to mitigate overfitting.

To further illustrate the impact of the number of latent factors (*k*) on predictive performance, we have now included an additional analysis (Supplementary Figures 20-23) evaluating model performance across different values of *k*, while keeping other model parameters fixed. When *k*=0 (i.e., no latent interactions are modeled), the predictive performance aligns with that of a standard Cox regression model, validating the baseline approach. As *k* increases, we observe an initial improvement in performance, particularly between *k*=0 and *k*=5. However, beyond *k*=5-10, performance plateaus in most cases, offering minimal additional gains. Moreover, larger *k* values increase the computational complexity, reinforcing the need for a conservative choice that balances predictive accuracy and efficiency.

We appreciate the opportunity to clarify our approach and have incorporated these additions to the revised manuscript to provide a clearer explanation for the selection of k and its implications for model performance (lines 457-470).

Line 363: thanks for providing the detailed derivation of gradient in equation 5-10; connected to line 488 – what are the procedures for selecting hyperparameters? Line 494 mentioned grid search but did not specify what are the criteria used (is it based on cross validation of likelihood? Prediction C-index?) (edit: this is explained in line 129-130)

Our response: We have now updated the description of the cross-validation and model evaluation procedures to align with the new training setup, where models are trained on individuals from England and Wales, and tested on individuals from Scotland. Additionally, we have explicitly clarified that regularization parameters are selected based on the maximum C-index across the validation folds. These updates are included both in the main text (lines 140-142) and in the Methods section (lines 607-617).

Line 107 – 109: Please provide more details on how prevalent cases were excluded. How did you exclude the prevalent diseases listed in ST2? Some of the diseases don't have primary care data available.

Our response: We have now added details on how the endpoints were constructed and prevalent cases excluded (lines 117-121, 575-584). We also note that primary care data are available only for a subset of participants. The endpoints were constructed including data from primary care (available for a subset of individuals), hospital episode statistics and death registries and self-reported diseases, using the first occurrences data field from UK Biobank (category 1712). For lung cancer, we additionally included data from the cancer registry. Prevalent cases were excluded based on their first recorded occurrence before the baseline assessment visit. Specifically, for each endpoint, individuals with a recorded diagnosis of the same condition prior to baseline were excluded.

Line 127: for some analysis (group 1, 2, and 4) it is also possible to run the analysis including all pairwise interaction terms. It is worth including a model with pairwise interaction terms as a comparison. This will potentially show under what predictor structure SurvivalFM shows best prediction power, in additional to sample size which is studied.

Our response:

We thank the reviewer for the great suggestion. In response, we have conducted additional analyses by training standard Cox regression models that explicitly enumerate all pairwise interaction terms. However, due to computational limitations, we were only able to run this analysis on the standard risk factor dataset ("Group 1"). Specifically, a single experiment with any of the other data types did not run with 32GB of memory, making it infeasible to run without substantial additional computational resources.

To ensure a fair comparison, we implemented a similar regularization strategy for the interaction terms in the standard Cox regression model as in survivalFM. Specifically, we allowed separate regularization strengths for the interaction and linear terms and optimized these hyperparameters using the same grid search procedure in cross-validation as applied to survivalFM. While survivalFM models interactions through a factorized parameterization (reducing the number of parameters and computational complexity), the standard Cox regression explicitly enumerates each pairwise interaction term.

The predictive performance Cox model incorporating all pairwise interactions was highly similar to that of survivalFM (Supplementary Figure 9). This key result highlights the ability of survivalFM to capture interaction effects while maintaining computational efficiency and a more compact parameterization.

This experiment also enabled us to compare the estimated effects (beta coefficients) for linear and interaction terms between the two models (Supplementary Figure 10). Despite differences in model structure and optimization algorithms, the estimated coefficients were highly similar between the two methods. To further confirm the comparability of the two approaches, we compared the predictions from both methods (Supplementary Figure 11). The predictions were highly concordant, with a correlation of 1 in 8 out of 9 scenarios and 0.99 in one scenario. This strong agreement shows that despite the methodological differences, both approaches yield highly comparable results.

We have now highlighted these results on lines 228-243 in the revised manuscript.

Line 212: connected to the main point – from the figure the improvement is small. It might be interesting to quantify what are the additional contributions of interaction for each of data modality.

Our response: We thank the reviewer for the suggestion. We have also addressed this in response to the main point above. Briefly, to better quantify the contribution of interaction terms and their added predictive value, we have incorporated an incremental R^2 measure (ΔR^2) based on Royston's R^2 (*Statistics in Medicine,* 2006). Royston's R^2 extends the concept of explained variation from linear regression to survival outcomes. This addition provides further assessment of the contribution of interactions for each data modality (Figure 1, Table 1, Supplementary Table 4, lines 153-160 and 315-322).

Line 247: Given the guideline recommended 10% risk threshold for QRISK3, for each model, it might be helpful to also report the observed risk of CVD event at 10 years (as well as other key time points) based on Kaplan-Meier analysis for patients with a predicted risk >=10% and those with a predicted risk <10%. This can provide an interpretable and clinically relevant summary of discrimination performance for each model which complement C-index and NRI reported.

Our response: Thank you for the great suggestion. We have now included a Kaplan-Meier analysis for each model (Figure 5c-d), assessing the observed risk of CVD events at various time points up to 10 years for patients with a predicted risk ≥10% and <10%.

As expected, the cumulative event rates at different time points are similar across models. This is by design, as we apply a 10% absolute risk cutoff and all models are well-calibrated, leading to comparable observed risks within the high- and low-risk groups. Specifically, the cumulative event rate is 15% in the high-risk group and 4% in the low-risk group across all models, with no statistically significant difference between the Kaplan-Meier curves.

However, these results show that survivalFM identifies a larger high-risk group at a comparable absolute risk of ≥10%, thereby capturing more events. By 10 years, survivalFM captures 844 events in the high-risk group, compared to 791 events captured by a standard Cox model (an increase of 6.7% in the captured events). We have now added these results to the manuscript (Figure 5c-d, lines 335-346).

Supplementary Figure 15: Figure legend should refer to a) and b) rather than c) and d).

Our response: Thank you for pointing this out, we have now corrected the legend accordingly.

Reviewer #2 (Remarks to the Author):

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Reviewer #2 (Remarks on code availability):

The survivalFM method is implemented in an R package, which can be a useful and user-friendly tool for interaction modeling in disease risk prediction. The Github page contains a README file with instructions on installation and example usage of the R package. I was able to install the R package and run the example code without issues.

Our response: We thank the reviewer for their time in co-reviewing our manuscript and for testing the installation and usage of our R package.

Reviewer #3 (Remarks to the Author):

The paper introduces survivalFM, a machine learning extension of the Cox proportional hazards model that incorporates pairwise interactions using a low-rank factorisation approach (from factorisation machines) to predict time-to-event outcomes. The authours evaluate survivalFM on 10-year disease outcomes using data from the UK Biobank, demonstrating improvements over traditional Cox models across multiple diseases and data modalities. The authours also specifically compare it to QRISK3 for cardiovascular risk prediction. The method shows particular strength in improving continuous NRI, while maintaining a level of interpretability.

The paper is well-written and demonstrates the potential of survivalFM for time-to-event prediction, but a few points would benefit from further clarification:

Interpretability of Latent Vectors: While it is true that the contribution of interaction terms to f(x) can be explicitly seen, could the authours clarify how the interaction coefficients derived from latent vectors in survivalFM differ in their clinical and statistical interpretation from traditional Cox models? In traditional Cox models, \beta_{ij} directly quantifies the strength and direction of the interaction between two covariates x_i and x_j, which is straightforward to interpret in a clinical context. However, in survivalFM, \langle p_i, p_j \rangle, the interaction derived from latent vectors, lacks the same direct interpretability, as it represents a more abstract relationship. How should clinicians understand these latent interactions in practice?

Our response:

We appreciate the reviewer's insightful questions about how interaction coefficients derived from latent vectors ($\langle \mathbf{p}_i, \mathbf{p}_j \rangle$) in *survivalFM* compare to the interaction coefficients (β_{ij}) in a standard Cox model. In standard Cox regression, β_{ij} directly quantifies the strength and direction of an interaction between two covariates x_i and x_j, which is often straightforward to interpret. In *survivalFM*, the interaction coefficient ($\beta_{ij} \approx \langle \mathbf{p}_i, \mathbf{p}_j \rangle$) serves an analogous role: it provides a single, interpretable coefficient reflecting the effect of covariate interactions on the hazard. The main difference is that *survivalFM* computes these interactions via factorized latent vectors rather than enumerating all pairwise interactions.

To further clarify these aspects, we have now conducted additional analyses by training standard Cox regression models that explicitly enumerate all pairwise interaction terms. Due to computational limitations, we were only able to run this analysis for the first input dataset consisting of standard risk factors (with the other input data types, a single experiment did not run with 32GB of memory and would have thus required substantial additional computational resources). Despite the differences in model parameterization and optimization algorithms, both methods exhibited similar improvements in predictive performance (Supplementary Figure 9). Moreover, when comparing the estimated coefficients (for both linear and interaction terms) between the Cox regression and survivalFM, we found that they were highly similar (Supplementary Figure 10). Furthermore, a comparison of the predictions from both models showed nearly perfect concordance with a correlation of 1 in 8 out of 9 scenarios and 0.99 in one scenario (Supplementary Figure 11). This strong agreement shows that despite the methodological differences, both approaches yield highly comparable results.

In practice, this means that the $\beta_{ij} \approx \langle \mathbf{p}_i, \mathbf{p}_j \rangle$ coefficients in *survivalFM* can be interpreted much like β_{ij} in a standard Cox model. Thus, *survivalFM* preserves the interpretability of interaction terms while offering a more compact representation that can alleviate the computational burden associated with estimating numerous interaction effects simultaneously. Although the factorized representation introduces an additional layer of abstraction (latent vectors), it does not diminish the interpretability of the resulting coefficients.

We have now clarified these aspects in the revised manuscript (lines 228-243). See also our response to reviewer #1 on a related suggestion on p. 6.

In Figure 5, the heatmap shows an interaction coefficient between mutually exclusive categories (e.g., heavy smoker:moderate smoker). How should this interaction be interpreted, and is it meaningful? Are there other unexpected interaction terms that might need further explanation?

Our response: We appreciate the reviewer's insightful comment. Indeed, our method can estimate interactions that do not co-occur in the training data. This arises from the low-rank factorization of the interaction matrix, which generalizes to feature interactions beyond those observed in the training data. Even if certain predictor pairs (e.g., "moderate smoker" and "heavy smoker") never appear together, the learned latent factors may still assign a nonzero interaction effect due to their generalized factor representations. Thus, while the method can learn meaningful interactions, it does not inherently respect real-world constraints, such as the mutual exclusivity of certain categorical variables, unless additional constraints are explicitly imposed. However, since such interactions never actually occur in real-world data, they do not affect the model predictions (as the corresponding product of one-hot-encoded features remains zero, i.e. that combination never appears in real inputs).

On the other hand, this ability to generalize to interactions not observed during training can also be beneficial, such as when working with categorical variables that have limited joint occurrences in the training data. We find this an interesting point and have now incorporated a discussion of these aspects into the revised manuscript (lines 471-483).

Selection of Latent Dimension (k): The authours use k=10 for the latent space dimension across all models. Could the rationale for this choice be explained? What guidelines could be provided for selecting k in different datasets or scenarios?

Our response:

We thank the reviewer for an insightful query regarding the selection of the latent space dimension *k*. Please see also our response to Reviewer #1 (p. 4-5) on a similar suggestion.

The number of latent factors (*k*) is a central hyperparameter in factorization-based models, as it determines the dimensionality of the latent space representing feature interactions. A higher *k* allows for greater expressiveness, while a lower *k* enforces a more compact representation. While cross-validation is one approach for tuning *k*, it becomes computationally prohibitive for large datasets. To manage the computational complexity, we adopted a pragmatic approach of selecting a moderate *k*=10, with L2 regularization applied separately to the latent factors to mitigate overfitting.

To further illustrate the impact of the number of latent factors (*k*) on predictive performance, we have now included an additional analysis (Supplementary Figures 20-23) evaluating model performance across different values of *k*, while keeping other model parameters fixed. When *k*=0 (i.e., no latent interactions are modeled), the predictive performance aligns with that of a standard Cox regression model, validating the baseline approach. As *k* increases, we observe an initial improvement in performance, particularly between *k*=0 and *k*=5. However, beyond *k*=5-10, the performance plateaus in most cases, offering minimal additional gains. Moreover, larger *k* values increase the computational complexity, reinforcing the need for a conservative choice that balances predictive accuracy and efficiency.

We appreciate the opportunity to clarify our approach and have incorporated these additions to the revised manuscript to provide a clearer explanation for the selection of k and its implications for model performance (lines 457-470).

List of Disease Outcomes: Could the authours explicitly list the 10 disease outcomes in the Methods section (I see they are buried in Supplementary Table 2)? This would help readers assess the diversity of outcomes and the fraction of improvements across scenarios.

Our response: Thank you for the suggestion. We have now explicitly listed the 10 disease outcomes in the Methods section (lines 579-582 and 586-587).

Model Composition: Why was a combined model using standard risk factors, clinical measures, metabolomics, and polygenic risk scores not included?

Our response:

Our initial approach focused on evaluating models that incorporate variables commonly measured together in clinical or research settings, reflecting a more feasible real-world prediction scenario. Additionally, analyzing different data types separately allowed us to compare the predictive contributions in each dataset. Furthermore, some data types (e.g., metabolomics) are only available for a subset of the cohort, which reduces the sample size for combined models.

However, we appreciate the reviewer's insightful suggestion and have now conducted an additional analysis combining all the input data types (standard risk factors, clinical measures, metabolomics, and polygenic risk scores) for individuals with measurements available across these modalities. The results are presented in Supplementary Figure 8 and discussed on lines 205-220 of the revised manuscript. These results show that while the improvements from modeling interactions were slightly attenuated in this combined model, likely due to the increased number of predictor variables, *survivalFM* continued to demonstrate improved risk prediction, particularly in terms of continuous net reclassification improvement (NRI).

Thank you for your consideration.

Reviewer #4 (Remarks to the Author):

Machine learning for comprehensive interaction modelling improves disease risk prediction in the UK Biobank

Heli Julkunen and Juho Rousu

The authors propose a model for all interactions of a Cox proportional hazards regression by projecting into a space of lower dimensions. Of course, this is also applicable to all linear models. As an example, the manuscript describes an application to the huge UK Biomank database combining clinical and genomic risk factors for a variety of diseases.

Comments

The present manuscript feels like an advertisement for the author's software package survivalFM. This manuscript would be an ideal candidate for a journal of statistical software. There is a lot of text devoted to the capabilities of survivalFM. If this is available in R then there are other, more suitable outlets for this manuscript

We respectfully disagree with the characterization of our manuscript as an advertisement for the software package. The primary focus of our manuscript is on the methodological innovation and scientific insights that our approach enables, rather than simply the software itself. The demonstration of improved predictive performance, while maintaining interpretability, is a key aspect of the cross-disciplinary research that illustrates the broader applicability of our method to, for instance, epidemiologists and clinical researchers.

First, we introduce a new methodological approach for predicting survival outcomes by leveraging factorized interaction terms, which, to our knowledge, has not been explored in this context before. This represents a substantive contribution to statistical and machine learning methodologies in survival analysis. Second, we apply this method in large-scale empirical analyses using the UK Biobank, evaluating its performance across multiple input datasets and disease outcomes. These analyses provide new insights into the role of interactions in risk prediction across various data modalities, offering findings that extend beyond the specific implementation.

Furthermore, Nature journals encourage the availability of software tools that facilitate reproducibility and broader adoption of new methods (https://www.nature.com/nature-portfolio/editorial-policies/reporting-standards). We view the R package release of our method as an asset that enables other researchers to apply, validate, and build upon our approach - something that should enhance rather than diminish the manuscript's contribution.

We believe that the methodological advances and scientific findings presented in this study align well with the scope of Nature Communications.

There is no strong connection to machine learning and this is not explained in any depth.

Our response:

We respectfully disagree with the assertion that there is no strong connection to machine learning. Our study employs a standard machine learning approach in which a model is trained on a training dataset to learn patterns and then applied to unseen test data to generate predictions. Specifically, our method learns from linear combinations of features along with interactions represented in a factorized form, similar to factorization machines - an established machine learning technique (see e.g. Rendle *IEEE International conference on data mining* (2010), Blondel et al. *NeurIPS* (2016)). However, in our study, this methodology is extended to accommodate time-to-event (survival) outcomes.

Machine learning is fundamentally about developing algorithms that learn from data and generalize to new observations, which is precisely the methodology applied in our study. If the reviewer has specific concerns regarding our implementation or its framing, we would be happy to address them.

There should be some guidance on how the dimensions of the p-parameters are chosen. In principal components analysis, for example, there is a scree plot to help us decide the appropriate dimension of the underlying latent measure. Similarly, there is no interpretation of the estimated p-vector.
For a given dimension of p, there is a good explanation (on page 10) on how p is estimated.

Our response:

We thank the reviewer for an insightful query regarding the selection of the dimension (rank *k*) of the vectors **p**. Please see also our responses to Reviewer #1 (p. 4-5) and Reviewer #3 (9-10) on similar suggestions.

The number of latent factors (*k*) is a crucial hyperparameter in factorization-based models, as it determines the dimensionality of the latent space representing feature interactions. A higher *k* allows for greater expressiveness, while a lower *k* enforces a more compact representation. While cross-validation is one approach for tuning *k*, it becomes computationally prohibitive for large datasets. To manage the computational complexity, we adopted a pragmatic approach of selecting a moderate *k*=10, with L2 regularization applied separately to the latent factors to mitigate overfitting.

To further illustrate the impact of the number of latent factors (*k*) on predictive performance, we have now included an additional analysis (Supplementary Figures 20-23) evaluating model performance across different values of *k*, while keeping other model parameters fixed. When *k*=0 (i.e., no latent interactions are modeled), the predictive performance aligns with that of a standard Cox regression model, validating the baseline approach. As *k* increases, we observe an initial improvement in performance, particularly between *k*=0 and *k*=5. However, beyond *k*=5-10, the performance plateaus in most cases, offering minimal additional gains. Moreover, larger *k* values increase the computational complexity, reinforcing the need for a conservative choice that balances predictive accuracy and efficiency.

We appreciate the opportunity to clarify our approach and have incorporated these additions to the revised manuscript to provide a clearer explanation for the selection of k and its implications for model performance (lines lines 457-470).

The choice of the two lambda parameters in (7) is very similar to the way penalized linear regression chooses the midground between lasso and ridge regression estimation.

Our response: As explained in the Methods section, we apply ridge regularization to both the linear part and latent vectors representing the interactions. We employ two separate regularization terms, i.e. one for the linear part and one for the latent vectors, to allow for different levels of penalization for these respective model components. Therefore, our approach does not combine ridge and lasso but rather applies two ridge penalties to the linear and interaction parameters. This is explained on lines 500-507 in the Methods section.

The frequent use of the word “prediction” is misleading. The manuscript is not predicting anything, but rather, seeking to build a mathematical model to explain what has already been observed.

Our response:

There seems to be a misunderstanding here.

First, our study follows a standard machine learning approach where a model is trained on a training set and subsequently applied to a test set to generate predictions of disease risk. These predictions are then evaluated for their accuracy in comparison to observed outcomes in the test set. This aligns with established machine learning terminology, where the term “prediction” is used to denote the application of a trained model to new or unseen data.

Second, the term “disease risk prediction” is widely used in epidemiology and public health research to describe statistical and computational approaches that estimate the risk of future disease occurrence (e.g., in prospective studies) based on a set of predictor variables.

Given the established usage of these terms in both fields, we see no reason to alter our wording.

NCOMMS-24-46967-A Point-by-point response to reviewer comments

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

The authors addressed my concerns well; I have the following additional (minor) comments:
1. Based on the updated results, the absolute gains in C-index and R squared are small in general from modeling interaction effects, while the predictive improvement mainly lies in NRI. Could the authors further comment on how this result aligns with previous studies modeling interaction terms in risk models, in terms of small non-additive effects compared to linear terms? The authors are suggested to clarify that including interaction terms provides modest improvement in prediction performance, which aligns with previous studies. The advantage in improving individual-level risk prediction despite small improvement at population level can be potentially emphasized.

Our response:

We thank the reviewer for the insightful comment. In response, we have revised the manuscript to explicitly acknowledge that the absolute improvements in global performance metrics (C-index, R^2) from modeling interaction effects are modest. We clarify that this finding aligns with prior studies, particularly in genetic risk prediction, where interaction effects have been shown to explain less variance than main effects. We also emphasize that, despite limited gains in global performance metrics, the inclusion of interaction terms can yield notable improvements in individual-level risk predictions, as reflected in NRI. These updates are included on lines 395-402 in the revised manuscript.

2. Figure 2 and line 164-166: since continuous NRI is not a proportion, might consider not using the percentage (as NRI can go up to 200%).

Our response:

We thank the reviewer for the suggestion. In response, we have revised the presentation of the continuous NRI throughout the manuscript to reflect absolute values rather than percentages. Specifically, we have updated all relevant figures and tables (Figure 2, Figure 3, Supplementary Figures 5–8, and Table 1), as well as the corresponding text (lines 179, 182, 190-193, 227-228, 340-342).

3. Figure 4: can you clarify if the grey line is Cox model with or without interaction terms? It is a fair comparison between Cox model with interaction term and survivalFM as survivalFM might require less samples to capture the interaction information. If the grey line is cox model with only main effect, it worth also putting the Cox model with all pairwise interactions in the figure.

Our response:

We thank the reviewer for this suggestion. We have now updated Figure 4 to include the Cox model with all pairwise interaction terms, as requested. In the revised figure, the grey line corresponds to the standard Cox model with main effects only, while an additional line now depicts the Cox model with interaction terms.

We would like to clarify that the primary purpose of this figure is not to compare SurvivalFM directly to the Cox model with interactions, but rather to illustrate how the ability to learn interaction effects depends on the size of the training dataset. Specifically, the goal is to show the incremental improvement achieved by models that capture interactions over models that include only main effects, across varying training dataset sizes. In this case, the dataset is sufficiently large relative to the number of parameters, such that the Cox model with all pairwise interactions serves as a reasonable upper bound on the performance achievable through explicit modeling of interaction effects.

We have now updated Figure 4 and lines 291-295 accordingly in the revised manuscript.

4. I noticed the first two numbers (and the corresponding lower bound) in row 1 of Table 1 are exactly the same.

Our response:

We appreciate the reviewer's careful attention to detail. The reason the first two numbers (and their corresponding lower bounds) in row 1 of Table 1 appear identical is that, after rounding to four decimal places (as presented in the table), the results from the standard Cox regression and the Cox regression model incorporating age interactions are numerically very close. Specifically, the C-index up to 16 decimal places for the standard Cox model is 0.7476032138010968 with a 95% confidence interval of (0.736937698294996, 0.7592745289480606), while the C-index for the Cox model with age interactions is 0.7475960083642162 with a confidence interval of (0.7369289725731945, 0.7587534571733379). Although these values differ beyond the fourth decimal place, rounding them to four decimals - as done for consistency and readability in the table - results in identical reported numbers.

5. Line 160 and Table S4: absolute R2 values range from 24.9% to 95.0%, which seems very high. Is this due to the specific properties of Royston's R2?

Our response:

We thank the reviewer for this observation. We believe the high R^2 values observed in some models are primarily attributable to the presence of strong prognostic factors, rather than to any specific mathematical property of Royston's R^2 measure.

The models yielding R^2 values above 90% include the prediction of chronic kidney disease using clinical chemistry biomarkers (R^2 95.02%) and metabolomics data (R^2 93.26%). Both models incorporate creatinine, an established and widely used biomarker of kidney function in clinical practice and in the formal diagnosis of kidney disease. Given the well-established association between creatinine levels and kidney outcomes, it is expected that such models would achieve high predictive performance for future incidence.

Similarly, the prediction model for type 2 diabetes based on clinical biochemistry and blood count data demonstrated a high R^2 value (R^2 94.39%). This model includes HbA1c, the gold-standard biomarker for diagnosing type 2 diabetes, which is also a strong predictor of future incidence. Therefore, the high R^2 values in these examples are consistent with the inclusion of robust, clinically validated prognostic markers.

Moreover, the high R^2 values are corroborated by correspondingly high C-indices (all exceeding 0.90) for these models, indicating excellent discrimination and reinforcing that the models are genuinely highly predictive. Thus, the observed high R^2 values reflect the strength of the predictors included in the models rather than an artifact of the mathematical properties of Royston's R^2.

6. Line 237 and Supplementary Figure 10: Would be better to say R > 0.99 instead of R = 1, which implies a perfect linear relationship.

Our response:

Thank you for the great suggestion. We have now revised the wording in line 246 and updated Supplementary Figure 10 to state "R > 0.99" instead of "R = 1".

Reviewer #2 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Our response:

We thank the reviewer for reviewing our manuscript.

Reviewer #3 (Remarks to the Author):

Thank you for your efforts in revising the manuscript. The updates have notably improved its clarity and rigour, and I now find the manuscript suitable for publication.

Our response:

We thank the reviewer for their positive assessment and are pleased that the revisions have improved our manuscript.

Reviewer #3 (Remarks on code availability):

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